

CircuLeak Industrial Carbon Audit

Executive Decision-Support Report — Facility: **Apex Metals & Casting Unit 4**
Audit Date: September 13, 2026 | Sector: Metals & Heavy Alloys

1. Executive Summary

An industrial carbon leak and circular diagnostic was executed for Apex Metals & Casting Unit 4. Analysis of 672 verified telemetry records indicates plant emissions of 20,076 kgCO₂e, operating at an intensity of 0.69 kgCO₂e per unit of production (-54.0% vs sector benchmark). 1 behavioral anomaly leaks were identified, with primary dissipation observed in Compressor 03 (Screw). Deployment of 3 prioritized circular interventions requires an estimated investment of INR 850,000, delivering INR 465,000 in recurring annual savings with a payback period of 1.8 years and reducing plant footprint by 49,100 kgCO₂e.

2. Facility Profile

Business Name	Apex Metals & Casting Unit 4	Sector	Metals & Heavy Alloys
Location	Vadodara Industrial Estate, Gujarat	Production Type	Alloy & Steel Fabrication
Production Volume	45,000 units	Operating Hours	24.0 hrs/day
Employees	280	Energy Sources	grid_electricity, natural_gas, diesel

3. Emission Overview

Metric	Value	Benchmark & Regulatory Context
Total Emissions	20,076.0 kgCO ₂ e (20.08 tCO ₂ e)	Total verified operational footprint
Emission Intensity	0.69 kgCO ₂ e/unit	Sector Avg: 1.5 (Best in Class (Top 10%))
CCTS Regulatory Threshold	1.3 kgCO ₂ e/unit	Status: Compliant
Total Electricity	11,785.0 kWh	Grid Factor: 0.716 kgCO ₂ e/kWh (India CEA v19)

4. Top Carbon Leaks (Structural Hotspots)

Rank	Equipment	Process	Emissions (kgCO ₂ e)	% Total
1	Boiler 02 (Steam)	Steam Generation	8,262.2	41.2%
2	Stenter Frame Oven	Thermal Processing	6,535.3	32.6%
3	Compressor 03 (Screw)	Compressed Air	3,470.5	17.3%
4	Primary Chiller Unit	Cooling & HVAC	1,808.0	9.0%

5. Behavioral Anomalies Detected

Equipment	Period	Risk	Baseline	Observed	Diagnosis
Compressor 03 (Screw)	22:00-06:00 (Off-Hours / Non-Production)	76.8	0.00	25.24	Compressor 03 (Screw) averages 25.2 kWh during non-production hours when baseline expectation is near zero.

6. Intervention Priority & Financial ROI

#	Intervention	CO2 Saved (kg)	Investment (INR)	Annual Savings (INR)	Payback	Feasibility
1	Boiler Flue Gas Waste Heat Recovery (Economizer)	18,500	INR 350,000	INR 180,000	1.94 yrs	High
2	Closed-Loop Pressurized Condensate Recovery System	16,000	INR 280,000	INR 150,000	1.87 yrs	High
3	Variable Frequency Drive (VFD) Retrofit on Air Compressor	14,600	INR 220,000	INR 135,000	1.63 yrs	High
4	Automated Boiler Standby Setback & Off-Hours Cutoff Controller	12,500	INR 55,000	INR 95,000	0.58 yrs	High
5	Ultrasonic Compressed Air Leak Audit & Sealing	9,200	INR 45,000	INR 120,000	0.38 yrs	High

7. Circularity Score & 5-Year Decarbonization Trajectory

Circularity Rating: Current Score = **78.1/100** (B). Projected Score post-interventions = **95.0/100**.

Year	Baseline (kgCO2e)	With Actions (kgCO2e)	Annual Reduction	Cumulative Savings
2026	20,076	7,027	13,049 kg	INR 196,000
2027	20,377	7,132	13,245 kg	INR 588,000
2028	20,683	7,239	13,444 kg	INR 1,148,000
2029	20,993	7,348	13,645 kg	INR 1,736,000
2030	21,308	7,458	13,850 kg	INR 2,352,000

8. Cryptographic Provenance & Official Auditor Sign-Off

OFFICIAL AUDITOR SIGN-OFF Lead Verifier: Dr. Rajesh K. Verma Credentials: Lead ISO 14064-3 Verifier & BEE Accredited Energy Auditor Accreditation ID: BEE-AEA/2026/0894 Accreditation Body: Bureau of Energy Efficiency (BEE) & GreenCarbon Council Audit Standard: ISO 14064-1:2018 / GHG Protocol Corporate Standard Status: Officially Certified & Verified Surveillance Cycle: Next due September 2027	CRYPTOGRAPHIC VERIFICATION SEAL SHA-256 Telemetry Fingerprint: af6c9da5cc65cfceb84b6565315d8b8a8f6c6065ccc1e981c22e39e717d2ed9b Ledger Integrity: Tamper-Evident SHA-256 Digest generated from raw telemetry and operational baseline data. Compliance Standing: Eligible for BEE Carbon Credit Trading Scheme (CCTS) & BRSR Core. Digital Seal: Verified by CircuLeak Enterprise Trust Engine v2.4
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